

Section 8.B. of the Self Study Report

Program Budget and Financial Support

The institutional budget for the Department of Chemical Engineering consists of two parts, personnel and the operating costs. The personnel costs are those for salaries of faculty, union staff members, academic professionals and teaching assistants. The budgeted amount for these salaries increases each year by an average percentage mandated by the University. These increases have so far been to retain qualified personnel in the department. The budget of the College is currently in part tied to the enrolment in Engineering and the tuition revenue generated by the enrolled students.

While the Department budget is not tied directly to enrolments, it is heavily influenced by it. The department has had a healthy increase in student enrolments in the last few years. This has resulted in healthy increases in our operating budgets, as well as allocations for Teaching Assistants (TAs) in the department. In contrast, the budget for the operating costs has not increased for the last few years and due to poor statewide economic conditions and has in fact decreased slightly in the last fiscal year. A fraction of the overhead funds available from research grants and contracts are returned to the Department as discretionary funds. These additional funds have therefore also been used to ensure that the program objectives are met. Some examples of how these overhead funds have been used in meeting program objectives are given below:

- (i) Augmenting the Teaching Assistants (TA) Budget in the Department: The allocated TA Budget is often not sufficient to cover all courses requiring TAs. The Department has routinely used overhead funds/gifts to hire additional TAs to maintain the quality of teachings mandated by our program objectives.
- (ii) Audio-Visual Resources: The department has used overhead funds to obtain such resources for the classrooms. The Department now has three projectors as well as associated desktop/laptop computers available for presentations, which is sufficient to cover our current and anticipated teaching needs.
- (iii) Updating Departmental Computing Services: Overhead funds have been used to obtain LCD monitors for our Computer Rooms as well as high performance computers that can be used efficiently with all the software packages our students are expected to use in their classroom activities (ASPEN, CHEMCAD. Etc.). In addition overhead funds have also been used to purchase site licenses for software needed by our students and faculty.

The current sources of financial support for the Department are the State of Illinois with additional funds available from gifts and overhead charges on research grants. State funding generally covers faculty and staff salary, as well as some additional non-permanent positions such as Teaching Assistants and part-time student workers. This portion of the budget is fixed each year by the University -- changes made are generally consistent with the support provided to the University by the State of Illinois. In addition the Department receives a portion of the Indirect Cost Recovery (ICR) from research grants housed in the department, or those in which the department faculty participate. Currently the University returns 50% of the overhead cost to the College of Engineering, and this is then divided equally between the College and the Department. The Department actively works with the Development Office of the College and University to raise gifts from alumni and local and national businesses with an interest in our Department and University. These gifts when they are discretionary are also used to fund special projects in the department including sometimes emergency repairs to equipment in the UG Laboratories that has not been budgeted. This year our development activities have resulted in one major gift and several smaller gifts.

While there have been state budget cuts over a period of time, alternate revenue streams have seen an increase driven by a steady increase in student enrollments and our international programs. This together with careful budgetary management in general by the Dean's office has allowed us to tide over the budget cutbacks and persist in providing a high quality education for the undergraduate students. Indeed, at the time of writing of this report, we are working with the Dean's Office to finalize an equipment procurement package totaling approximately \$250K. This will be directed toward upgrading the Unit Operations Laboratory (CHE 381, 382) and enhancing classroom and learning technology. College alumni donor funding for the recently inaugurated Student Collaboratorium workspace is another example of support for our program. Strong support enables our effectiveness in teaching and learning. The College conducts surveys of teaching and advising every semester, and the Chemical Engineering department usually ranks the highest in these surveys. Another measure of the quality of instruction we have provided to our students is the Annual Engineering Expo. This year as in the past years, our students have won a disproportionately higher share of awards. As an example, in 2014 Chemical Engineering students won the first prize in the Chemical Processes and Chemical Production Methods and Facilities categories.